

How to write a good research paper

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For Future
Reference!

This lecture is a resource on how to write a good paper/report, which you can use for any upcoming assignments.

Goal of this lecture

General Tips and Structure

General writing tips



- Use **simple** sentence structure, don't hide your point behind complex formulations.
- Use **active** formulations (avoid passive formulations), use **first person** pronouns.
 - there is a balance between these three, not every sentence should start with "We describe... We implement...", but its usage is generally accepted and welcome, especially to simplify sentences.
- Avoid **repetition of words** in the same sentence (does not apply to specific names of methods etc.).
- Resources:
 - Synonyms: <https://www.thesaurus.com/>
 - Collocations: <https://www.freecollocation.com/>

Example/Exercise (recap)

- ☒  The results of [..] are presented in Table 1.
- ☒  Table 1 presents the results of [..].

- ☒  Algorithm 1 outperformed the baseline.
- ☒  The baseline was outperformed by Algorithm 1.

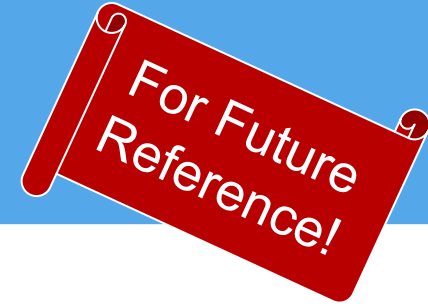
- ☒  Algorithm 1 was introduced by Miller et al. [1].
- ☒  Miller et al. [1] introduced Algorithm 1.

Examples of Collocations



- **Adverb + adjective:**
We are fully aware of the consequences.
- **Adjective + noun:**
He has carried out extensive research into renewable sources.
- **Noun + noun:**
He has carried out extensive research into renewable sources.
- **Noun + verb:**
He has carried out extensive research into renewable sources.
- **Verb + noun:**
He gives a presentation about his work.
- **Verb + expression with preposition:**
I spent two year carefully researching into his background.
- **Verb + adverb:**
I spent two year carefully researching into his background.

Choosing the right verb tense



Use **past tense** to present those events that have happened.

- describing previous work of other people. (related work)
- describe specific findings (results) or events/choices (limitations)

Use **present tense** to indicate events that are going on.

- describing findings that still hold true (motivation/setting)
- describing a method/model as the approach is timeless and doesn't change just because you have executed it. (approach/methodology)

Use **future tense** to indicate events that will happen in the future. → describing future work, this tense is only used in the future work section or when specifically indicating that something can be done in the future, not in the presented work. (future work)

Example/Exercise (recap)

Our approach [...] the recalculated features to estimate ...

- ☒ uses (present)
- ☒ used (past)
- ☒ will use (future)

We [...] the pizza ontology for the evaluation because ...

- ☒ choose (present)
- ☒ chose (past)
- ☒ will choose (future)

The number of participants [...] the generalizability of our findings.

- ☒ limits (present)
- ☒ limited (past)
- ☒ will limit (future)

Structuring a general paragraph



Each paragraph must be:

- **Unified:** all the sentences relate to one central point or idea.
- **Coherent:** the sentences are logically organized and clearly connected.
- **Relevant:** the paragraph supports the overall theme and purpose of the paper.

Structure of a paragraph (general):

- Topic sentence
- Evidence taken from sources (and citation)
- Stitching that effectively links to those sources to one another (and citation)
- Conclusion

Essay Writing Guide



Guide, shared on Canvas, on the **process of writing** a great essay by Jordan B. Paterson!

Careful, not all tips translate one to one for scientific writing, but the process is a **valuable tool** outlined / summarized briefly in the next slide.

10 parts

1. Introduction (What is an essay, why bother..., a note on technology, use of time)
2. Levels of Resolution (Words, sentences, paragraphs, more)
3. The topic and the reading list
4. Outline
5. Paragraphs
6. Editing and arranging of sentences within paragraphs
7. Re-ordering the paragraphs
8. Generating a new outline
9. Repeat
10. References and Bibliography

**If you can't write, it is because you have nothing to say.
You have no ideas. In such a situation, don't pride
yourself on your writer's block. Read something. If that
doesn't work, read something else - maybe something
better. Repeat until the problem is solved.**

Jordan B. Paterson

A note on taking notes



- Read a bit, **write down** what you've learned, or any question that came up
- The most important part is the recreation of what you have read in your **own language**
 - It goes beyond "use or own words"
 - Form of Dialog with the author "this is what I understand you are saying".
 - **DO NOT COPY THE ORIGINAL TEXT!**
- Take about 3x as many notes as you would need for your (finished) writing.
- If you find note taking in this manner difficult, try this approach:
 - Read a paragraph.
 - Look away.
 - Then say to yourself, out loud, what the paragraph meant.
 - Listen to what you said.
 - Quickly write it down.

Don't bother yourself with trying to write the best sentence from the beginning, that's what editing is for. In the beginning its about putting something on paper. Perfect doesn't exist anyway.

Jordan B. Paterson

Using Generative AI

Every course has different rules. The rules presented here **only apply to the Introduction to AI course!**

You are allowed to use ChatGPT as a **writing help** for:

- Fixing grammar mistakes
- Rephrasing
- Shortening

Ideas need to come from you!

General Structure



1. Introduction
2. Related Work
3. Approach
4. Evaluation (including Data, can be combined with Approach)
5. Results
6. Discussion (can be combined with Results)
7. Limitations and Future Work
8. Conclusions

Introduction

1. Introduction



This is all in the same section, each can have its own paragraph or two, sometimes two of them can be in one but all are present:

1. **Setting**: What is the domain, high level task, why do we care about the domain/task
2. **Problem and its motivation**: What is the problem in the domain/task, why do we care about it, why does it need to be solved?
3. **Problem Statement** and Research Question
4. **Approach**: how will you solve the problem
5. **Evaluation**: how will you evaluate that you are solving the problem
6. Findings: what did you learn from the evaluation
7. Explanation of paper/report structure.

Example Paper

<https://doi.org/10.1145/1273445.1273458>



“How to read a paper”

1. **Setting**: What is the domain, high level task, why do we care about the domain/task
2. **Problem and its motivation**: What is the problem in the domain/task, why do we care about it, why does it need to be solved?
3. **Problem Statement** and Research Question
4. **Approach**: how will you solve the problem
5. **Evaluation**: how will you evaluate that you are solving the problem
6. Findings: what did you learn from the evaluation
7. Explanation of paper/report structure.

Researchers must read papers for several reasons: to review them for a conference or a class, to keep current in their field, or for a literature survey of a new field. A typical researcher will likely spend hundreds of hours every year reading papers.

Learning to efficiently read a paper is a critical but rarely taught skill. Beginning graduate students, therefore, must learn on their own using trial and error. Students waste much effort in the process and are frequently driven to frustration.

For many years I have used a simple approach to efficiently read papers. This paper describes the ‘three-pass’ approach and its use in doing a literature survey.

Related Work

Write your own Related Work



1. Every time you read a paper, write a (very) short summary of the paper
2. Check references of papers you read to search for other papers to read.
3. After a first brief pass of the title, abstract and introduction, answer the question “how does this relate to my work?”
4. There will be different answers to the question above - this will give the related work “themes”, that will make separate paragraphs
5. Start with “headers” for the themes and put citations with the very short summaries under them to start off.

Paragraph structure in Related Work Section



Remember the general paragraph structure:

- Topic sentence
- Evidence taken from sources (and citation)
- Stitching that effectively links to those sources to one another (and citation)
- Conclusion

Applied to Related Work paragraph:

- Topic sentence = Introductory sentence (what kind of related work are you about to present, what do they have in common on a high level)
- Evidence taken from source:
 - Paper 1 description. (1-3 sentences max.)
 - Paper 2 description. (1-3 sentences max.)
 - More paper descriptions...
- Conclusion = How is this work different from the above mentioned papers.

Example: Toward Measuring [..]

- Introductory sentence
- Paper 1 description.
- Paper 2 description.
- More paper descriptions...
- How is this work different from the above mentioned papers.

A well-known consequence of evolution is semantic drift or semantic impact. SemaDrift [22] is a tool to calculate various semantic drift measures between versions of ontologies. It applies different methods of calculation and distinguishes between an exact, inexact, and hybrid ontology matching approach. In OntoDrift [5], the authors introduce an update to semantic drift measures used in SemaDrift for a better global comparison. In contrast to these tools, we compare the impact of ontology change in the context of embeddings, which can then be used in multiple more specific tasks like link prediction, classification, or recommender systems, instead of the ontologies directly.

There are other studies that directly investigate the effect of ontology evolution on a specific task. Gonçalves et al. [8] define a categorization of changes based on entailment impact. For the classification of a change, they investigate if it influences the set of entailed axioms in the next version. Gross et al. [10] examine how the changes in an ontology impact previously conducted functional analysis. They define an impact measure based on stability of concepts over time and investigate the results of enrichment analysis using real and synthetic data to prove the effectiveness of the measure. Gottron and Gottron [9] also investigate the impact of KG evolution using LOD. They implement twelve different indexing methods and evaluate how the indices are affected by the evolution of the data using three different measures. Osborne and Motta [14] present the pragmatic ontology evolution, in which they analyze the selection of concepts for a new version by evaluating the performance of four different tasks. Our work, in contrast, focuses on the direct comparison of embedding models.

How to cite



Citation at the beginning of the sentence:

- In **OntoDrift [5]**, the authors introduce an update to semantic drift measures used in SemaDrift for a better global comparison.
- **Wegmann et al. [29]** focus on language models and investigate different types of semantic shifts in a specific language.

Citation at the end of the sentence:

- A visual approach has emerged implementing comparison methods and presenting neighbourhood distributions **[2]**.
- Looking beyond graph embeddings, there exist approaches to compare word embedding models in language processing **[1, 27, 29]**.

~~(\cite{key1}\cite{key2})~~ → \cite{key1, key2, key3})

Methodology, Results and Discussion

Methodology, Approach, Evaluation



These sections, no matter how divided, iterate facts, not opinions.

- Written in an objective way
- Enough detail for reproducibility
- Justification of choices, assumptions

It needs to include (examples):

- Describe the data used
- Describe the data proportion
- Describe the research protocol
- Explain how/what is measured, what calculations are made, statistical tests etc.

Results



Results presents and describes the output of the evaluation (experiments, or other, depending on the research methodology).

- **Objectively** report your evaluation output.
 - Answer related questions/hypotheses
 - Make us of **visual** presentation of numbers, especially for comparisons.
 - Every visual representation of results has to be **explained** in the text (in detail)
- x Not speculate** about the meaning or attempt to answer the main research question.

Discussion - DOs

For Future
Reference!



Discussion - DON'Ts



- X Don't** introduce new results:
Only discuss what you have already reported in the results section.
- X Don't** make inflated claims:
Avoid overinterpretation and speculation that isn't directly supported by your data and results.
- X Don't** undermine your research:
The discussion of limitations should aim to strengthen your credibility, not emphasize weaknesses or failures.

Limitations, Future Work, Conclusions

Limitations & Future Work (1 section)



- Not generalizable because
 - Only x number of methods tested
 - Only x different number of datasets used
 - Only x domain examined
 - In the future, to make findings generalizable, add methods that do y, also use datasets from z, applied to other domains such as ...
- Limitation of the method used
 - Doesn't handle certain types of outliers
 - Use a different method for those cases in the future
 - Only knowledge representation approaches
 - In the future also ML approaches (embeddings, (L)LMs, etc.)

Conclusion



- **Restate** the problem statement addressed in the paper
 - Motivation, problem, solution
- **Summarize** your overall arguments or findings
 - From results, and discussion, as well as limitations
- Suggest the **key takeaways** from your paper
 - How do the findings influence the overall field
 - Why was this work so important for the field
 - etc.

Example Conclusion: Toward Measuring [..]

The impact of ontology evolution is getting more attention as more ontologies are shared across the web and used in various applications. It is essential to consider the consequences of the growing and ever-changing knowledge that we, as researchers, capture in ontologies and use further [17, 29]. The consequences on the embedding have not been addressed before.

We found [...] We answer our research question by defining an impact measure called Embedding Change Impact (ERI). ERI builds on similarity measures and overcomes their shortcomings. ERI is orthogonal to similarity measures. It captures impact of changes but does not give indication on stability of embedding methods.

To evaluate ERI, we investigated [...]. Therefore, we conclude that the proposed impact measure is valid for the application of embedding model comparison. We show that ERI overcomes the shortcomings of similarity measures—stochasticity of embeddings and disregard for changes to the ontology—but also has limitations.

This research is highly relevant, because it allows future investigation into the impact of ontology evolution on embeddings. With ERI, we can now investigate the learning and prediction of impact and informing ontology users about the necessity of recalculation as proposed by Wegmann et al. [29]. Approximating and/or predicting impact, like attempted by [18] but with similarity measures, would signal to ontology users if they need to update their current application and adapt to the new version. If the changes between versions were not enough to warrant the update, resources can be saved and used for the next update of the ontology and application.

Depending on the application, only recalculating every other time could already be highly beneficial, if previously an update was made every hour. Hence, with ERI, we enable the research into this area, which could benefit many engineers, applications, and companies.

- Restate
- Summarize
- **Key takeaways**
 - a. How do the findings influence the overall field
 - b. Why was this work so important for the field
 - c. etc.

References

Formatting References



- Make use of a bibtex file, don't format your references by yourself
- If you used Zotero, once you are done with adding references, download the file to your computer and upload it into Overleaf, so that you can edit it.
- Be consistent about the information you have in your references, e.g. use full names of journals for every publication, don't abbreviate one and leave others full.
- Do this at the very end of formatting, once you are done with everything else.
- Make sure every reference has the following information:
 - Author
 - Title
 - Year
 - DOI (remove SSN or other identifier), none if the publication doesn't have an identifier
 - Publication information - this differs between `@article` (journal) and `@inproceedings` (conference)
 - Journal: Journal name, publisher, edition, page numbers
 - Inproceedings: Proceedingsname (name of the conference), page numbers, publisher

Lessons learned

Take Home messages

These slides are a **resource** for all your studies, use them!

More specific:

- Active over passive verb form.
- Perfect doesn't exist.
- Note taking.
- Paragraph structure.
- Structure of a paper.
- Section specific dos and don'ts.

The Practical: LaTeX

Overleaf

The screenshot displays the Overleaf online LaTeX editor interface. The top navigation bar includes a 'Menu' icon, an 'Upgrade' button, the text 'Springer Conference Proceedings Template Updated 2022-01-12', a 'Review' button, and icons for 'Share', 'Submit', 'History', 'Layout', and 'Chat'. Below this, a secondary bar shows 'Source' (selected), 'Source (legacy)', and 'Rich Text' tabs, along with a 'Recompile' button and icons for file management.

The left sidebar contains a file explorer with the following items: `fig1.eps`, `history.txt`, `llncs.cls`, `llncsdoc.pdf`, `readme.txt`, `samplepaper.tex` (highlighted), and `splncs04.bst`. Below the file explorer is a 'File outline' section showing a tree structure: 'First Section' > 'A Subsection Sample' > 'Sample Heading ...' > 'Sample Head...' > 'Acknowledgeme...'.

The main editor area shows the LaTeX source code for `samplepaper.tex`. The code includes comments in German and LaTeX commands for document class, packages, fonts, and document structure. The right pane displays the compiled PDF output, which is a Springer conference proceedings template. The PDF content includes a title page with 'Contribution Title', author information, an abstract, keywords, a table of contents, and a list of references.

The PDF content details:

- Title:** Contribution Title^{*}
- Authors:** First Author¹⁰⁰⁰⁻¹¹¹¹⁻²²²²⁻³³³³, Second Author¹¹¹¹⁻²²²²⁻³³³³⁻⁴⁴⁴⁴, and Third Author²²²²⁻⁻³³³³⁻⁴⁴⁴⁴⁻⁵⁵⁵⁵
- Footnotes:**
 - ¹ Princeton University, Princeton NJ 08544, USA
 - ² Springer Heidelberg, Tiergartenstr. 17, 69121 Heidelberg, Germany
 - ³ ABC Institute, Super-Kale-University Heidelberg, Heidelberg, Germany
- Abstract:** The abstract should briefly summarize the contents of the paper in 150-250 words.
- Keywords:** First keyword - Second keyword - Another keyword.
- 1 First Section**
 - 1.1 A Subsection Sample**

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.
 - Sample Heading (Third Level)** Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.
 - Sample Heading (Fourth Level)** The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.
- Table 1:** Table captions should be placed above the tables.
- Table:**

Heading level	Example	Page size and style
Title (unnumbered)	Lecture Notes	24 point, bold
Section heading	1 Introduction	12 point, bold
Section heading	2.1 Printing Area	12 point, bold
Section heading	2.1.1 Printing Area	12 point, bold
Section heading	2.1.1.1 Printing Area	12 point, bold
Section heading	2.1.1.1.1 Printing Area	12 point, bold

^{*} Requested by organization x.

2 F. Author et al.

Displayed equations are centered and set on a separate line.

$$x + y = z \quad (1)$$

Please try to avoid centered images for line-art diagrams and schemes. Whenever possible, use vector graphics instead (see Fig. 1).

Fig. 1

- Formatting done “automatically” with a compiler that generates the PDF.
- Sections using `\section{..}`, `\subsection{..}`, etc.
- Images, tables, code → see Overleaf Help pages
- Follow the template and its explanations for correct formatting.

Cross-referencing: `\label{..}` and `\ref{..}`

For Future Reference!

- `\label{key}` remembers the last “number”, e.g. section, figure, etc.
- `\ref{key}` prints that number, `\Cref{key}` prints the kind and number, e.g. Section 2.3 or Figure 2, etc. (for this command you need to add `\usepackage{cleverref}` before `\begin{document}`)
- Example with a figure
- This works also for sections or tables.

```
\section{Introduction} \label{introduction}
This is an introductory paragraph with some dummy text. This section will be
referenced later.
```

```
\begin{figure}[h]
\centering
\includegraphics[width=0.3\linewidth]{overleaf-logo}
\caption{This image will be referenced below.}
\label{fig:leaf}
\end{figure}
```

```
You can reference images; for instance, Figure \ref{fig:leaf} shows the
\textit{Overleaf} logo.
```

Footnotes and when to use them



- `\footnote{..}` right behind text, without a space.
- To signal “references” which are not proper scientific articles like blogs, pretrained models, code packages.
 - Careful, some pretrained models and code packages have publications associated with them!
- Put `\url{..}` when its a link, add the date when accessed.
- With code packages, if a proper release, add the release number.
- Normally we use a footnote to share code
 - you can put your code in a public GitHub and put the link into your thesis. E.g. All code and data available online`\footnote{\url{www.github.com/...}}` .

Format of paragraphs



- The template does the formatting for you, **don't fight it!**
- First paragraph after section headings is not indented, subsequent paragraphs are indented.
- Space between paragraphs is dictated by the template it will add more space to fill the page.
- Don't place two section commands after each other without any text between them. (not like this!)

1 First Section

1.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.